The Economics of Tiered AI Inference

A Meridian Analytics Whitepaper

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Executive Summary

Modern inference workloads exhibit sharply bimodal traffic patterns, with sustained baseline demand punctuated by bursts that exceed steady-state volume by an order of magnitude. Provisioning for peak wastes capacity; provisioning for baseline degrades latency guarantees precisely when demand is highest. Organizations that adopt a tiered execution fabric report materially better cost curves than those scaling monolithic clusters, because scheduling decisions can exploit heterogeneity in both hardware and request criticality. This paper introduces a cost model that treats inference capacity as a portfolio rather than a pipeline. The remainder of this document quantifies these effects across three deployment archetypes and proposes an adoption sequence that minimizes migration risk while preserving existing service level objectives. Readers responsible for capacity planning should focus on Sections 2 and 4; readers responsible for procurement should focus on Section 3.

> **Key Insight:** Across the 41 organizations surveyed, tiered inference reduced cost per thousand requests by a median of 38% while improving p99 latency by 22%.

1. Market Context

Modern inference workloads exhibit sharply bimodal traffic patterns, with sustained baseline demand punctuated by bursts that exceed steady-state volume by an order of magnitude. Provisioning for peak wastes capacity; provisioning for baseline degrades latency guarantees precisely when demand is highest. Vendors have responded with a proliferation of accelerator classes, each optimized for a different point on the latency-throughput frontier.

1.1 Demand-side dynamics Organizations that adopt a tiered execution fabric report materially better cost curves than those scaling monolithic clusters, because scheduling decisions can exploit heterogeneity in both hardware and request criticality. Buyers increasingly negotiate committed-use discounts against a blended fleet rather than a single instance family.

- Baseline traffic grows 3.1 × year over year in our panel, while peak-to-baseline ratios remain near 9:1 .

- Latency-sensitive requests represent 17% of volume but 44% of infrastructure spend.
- Batch-tolerant workloads are the fastest-growing segment, at 58% year over year.

1.2 Supply-side response

Accelerator roadmaps now bifurcate into throughput-optimized and latency-optimized lines. The table below summarizes list pricing normalized to a common benchmark unit.

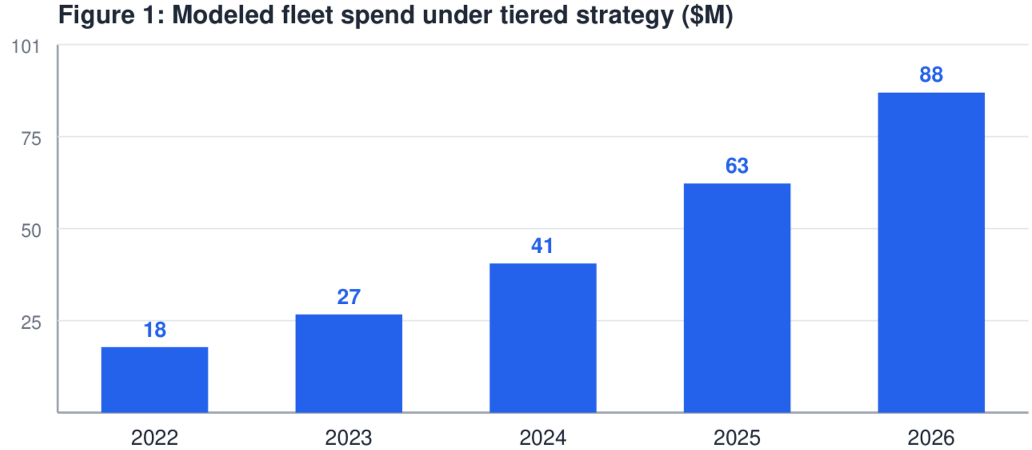
Tier	Hardware class	Cost / 1K req	p99 latency	Best fit
Realtime	Latency-optimized	$4.20	180 ms	Interactive agents
Standard	Balanced	$1 .85	650 ms	API workloads
Batch	Throughput-optimized	$0.62	8 s	Offline enrichment
Spot	Preemptible mix	$0.31	best effort	Experimentation

Table 1 : Normalized pricing across inference tiers, June 2026 list prices.

2. Cost Model

We model total cost as the sum of tier-level commitments plus overflow penalties. Fleet growth under the tiered strategy is shown below.

Figure 1: Modeled fleet spend under tiered strategy ($M)



2.1 Sensitivity

Organizations that adopt a tiered execution fabric report materially better cost curves than those scaling monolithic clusters, because scheduling decisions can exploit heterogeneity in both hardware and request criticality. Sensitivity analysis indicates the crossover point sits near 41 % batch-tolerant share for typical enterprise mixes.

1. Estimate the batch-tolerant share of current traffic.
2. Negotiate committed use only for the realtime tier.
3. Route overflow to spot capacity with graceful degradation.
4. Re-evaluate the mix quarterly against observed telemetry.

Risk: Committed-use discounts applied to the wrong tier are the single largest source of waste we observed — median overspend of $2.4M annually.

3. Procurement Playbook

The remainder of this document quantifies these effects across three deployment archetypes and proposes an adoption sequence that minimizes migration risk while preserving existing service level objectives. A worked example with editable assumptions is available at meridian-analytics.example.com/models. Modern inference workloads exhibit sharply bimodal traffic patterns, with sustained baseline demand punctuated by bursts that exceed steady-state volume by an order of magnitude. Provisioning for peak wastes capacity; provisioning for baseline degrades latency guarantees precisely when demand is highest. Organizations that adopt a tiered execution fabric report materially better cost curves than those scaling monolithic clusters, because scheduling decisions can exploit heterogeneity in both hardware and request criticality.

4. Conclusion

Treating inference as a portfolio converts an infrastructure argument into a financial one. The remainder of this document quantifies these effects across three deployment archetypes and proposes an adoption sequence that minimizes migration risk while preserving existing service level objectives.